FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: fault
Model status: BOUNDED_SINGLE_RADIAL_SOURCE_SCREEN
Captured UTC: 2026-10-26T06:04:36.511Z
Warning: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certification or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: nominalLLV | Entered value: 13800
Input path: voltageFactor | Entered value: 1.1
Input path: frequencyHz | Entered value: 60
Input path: seriesImpedances[0].id | Entered value: utility-positive-Z
Input path: seriesImpedances[0].rOhm | Entered value: 0.12
Input path: seriesImpedances[0].xOhm | Entered value: 0.94
Input path: seriesImpedances[1].id | Entered value: feeder-at-fault-voltage-base
Input path: seriesImpedances[1].rOhm | Entered value: 0.08
Input path: seriesImpedances[1].xOhm | Entered value: 0.21
Input path: dcTimesS[0] | Entered value: 0
Input path: dcTimesS[1] | Entered value: 0.01
Input path: dcTimesS[2] | Entered value: 0.03
Input path: dcTimesS[3] | Entered value: 0.05

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: BOUNDED_SINGLE_RADIAL_SOURCE_SCREEN
Calculated output path: method | Value: IEC 60909-0:2026 single equivalent-voltage-source, symmetrical initial three-phase current and radial equivalent-R/X peak screening; not a full IEC 60909 calculation. No distributed source, inverter, generator/motor decrement, meshed network, corrections, min/max case selection, fault impedance or interrupting duty certification.
Calculated output path: nominalLLV | Value: 13800
Calculated output path: voltageFactor | Value: 1.1
Calculated output path: frequencyHz | Value: 60
Calculated output path: totalROhm | Value: 0.2
Calculated output path: totalXOhm | Value: 1.15
Calculated output path: impedanceMagnitudeOhm | Value: 1.167261752992875
Calculated output path: rxRatio | Value: 0.1739130434782609
Calculated output path: initialSymmetricalA | Value: 7508.321988471781
Calculated output path: initialSymmetricalKA | Value: 7.508321988471781
Calculated output path: kappa | Value: 1.6016177478009073
Calculated output path: peakA | Value: 17006.57110480697
Calculated output path: peakKA | Value: 17.006571104806973
Calculated output path: initialShortCircuitMVA | Value: 179.46617325795194
Calculated output path: decayingDC[0].seconds | Value: 0
Calculated output path: decayingDC[0].dcOffsetMagnitudeA | Value: 10618.37078676092
Calculated output path: decayingDC[1].seconds | Value: 0.01
Calculated output path: decayingDC[1].dcOffsetMagnitudeA | Value: 5512.117587199794
Calculated output path: decayingDC[2].seconds | Value: 0.03
Calculated output path: decayingDC[2].dcOffsetMagnitudeA | Value: 1485.3870990088822
Calculated output path: decayingDC[3].seconds | Value: 0.05
Calculated output path: decayingDC[3].dcOffsetMagnitudeA | Value: 400.27717097792913
Calculated output path: missingForFullStandard[0] | Value: IEC 60909 source and equipment factors
Calculated output path: missingForFullStandard[1] | Value: full positive, negative and zero sequence network

Calculated output path: missingForFullStandard[2] | Value: voltage-factor choice
Calculated output path: missingForFullStandard[3] | Value: all DER and motor contributions
Calculated output path: missingForFullStandard[4] | Value: near-generator breaking-time decrement
Calculated output path: missingForFullStandard[5] | Value: breaker making and breaking duty
Calculated output path: missingForFullStandard[6] | Value: meshed peak factors
Calculated output path: missingForFullStandard[7] | Value: fault location/case validation

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: fault
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:36.511Z